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MANUFACTURING METHOD OF ELECTRONIC PACKAGE

Abstract

A manufacturing method of an electronic package is provided, which includes: adhering a packaging module to a surface of a carrier through a first adhesive layer, in which the packaging module has a plurality of packaging units jointly defined by first and second cutting paths; performing a pre-cutting process to cut along each of the first cutting paths; disposing at least one reinforcement structure on the surface of the first adhesive layer; adhering a plurality of optoelectronic components to the packaging units through a second adhesive layer; electrically connecting each of the optoelectronic components to a corresponding one of the packaging units; performing a singulation process to cut down the packaging module along the second cutting paths; and removing the reinforcing structure, the first adhesive layer and the carrier to form a plurality of the electronic packages.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] The present application is based upon and claims the right of priority to TW Patent Application No. 113107065, filed Feb. 27, 2024, the disclosure of which is hereby incorporated by reference herein in its entirety for all purposes.

BACKGROUND

1. Technical Field

[0002] The present disclosure relates to a manufacturing method of an electronic package, and more particularly, to a manufacturing method of an electronic package having optoelectronic component.

2. Description of Related Art

[0003] With the vigorous development of the electronics industry, it has become a trend for electronic products to develop towards multi-function and high performance. In particular, the demand for data transmission has emerged largely, causing the industry to begin to use “light” instead of “electricity” as the carrier for data transmission, thereby to improve the transmission capacity, efficiency and/or distance, and to reduce the energy consumption during the transmission process. Under this background, the Co-Packaged Optical structure and the Co-Packaged Optics technology that include both the optical (photonic) components and optoelectronic components have become the trend of development for future semiconductor and packaging technologies.

[0004] FIG. **1**A to FIG. **1**C are schematic views illustrating a manufacturing process of a conventional semiconductor package **1**, and FIG. **1**B-**2** is a schematic partial top view of FIG. **1**B-**1**. A fan-out module **10** is firstly adhered to a carrier board **12** through an adhesive layer **11**. An electronic IC (“EIC”) component **104** is embedded in the molding compound (“M/C” in short) **103** of the fan-out module **10**. Then a first cutting is performed along the longitudinal cutting paths to remove a part of the fan-out module **10** located on the longitudinal cutting paths to form a plurality of longitudinal grooves **101**, such that the adhesive layer **11** underneath is exposed from the bottom surfaces of the longitudinal grooves **101**. After that, a plurality of photonic IC (PIC) components **13** are disposed over the fan-out module **10**, and the PIC components **13** are coupled to the fan-out module **10** through a thermal compression process. After all the PIC components **13** have been connected to the fan-out module **10** firmly, a second cutting is performed along each of the transverse cutting paths shown in FIG. **1**B-**2**. Finally, the carrier board **12** and the adhesive layer **11** are removed, so that a plurality of semiconductor packages **1** shown in FIG. **1**C can be obtained.

[0005] However, in the aforementioned conventional processes, the heat imposed during performing the thermal compression process to the PIC component **13** will dissipate to the adhesive layer **11** and causes the adhesive layer to soften and even melt, and the downward pressing force will make the remaining fan-out module **10** compress the adhesive layer **11** after the first cutting, causing the adhesive material of the adhesive layer **11** within the longitudinal grooves **101** to deform to result in bulge **11***a*, as shown in FIG. **1**B-**1**. Then, the unremoved fan-out module **10** on the transverse cutting paths **102** and/or in the nearby area will be pushed by the deformation of the adhesive layer **11** or the bulge **11***a*, causing fracture (as shown in FIG. **1**B-**2**) of the fan-out module **10** or other structural damage to occur on those locations or areas. Therefore, the final product would become a defective product due to the structural issues. In addition, the adhesive layer **11** that is softened or melted by heat will also make the fan-out module **10** above it easy to slide and deviate due to the downward pressing force during the thermal pressing process, resulting in deviations in the subsequent manufacturing process, such as singulation. All these problems will adversely affect the yield rate of the above-mentioned conventional manufacturing process.

[0006] Therefore, how to overcome the above-mentioned problems of conventional techniques has become a critical issue to be solved.

SUMMARY

[0007] In view of the aforementioned shortcomings of the prior art, the present disclosure provides a manufacturing method of an electronic package, comprising: adhering a packaging module onto a surface of a carrier through a first adhesive layer, wherein a plurality of first cutting paths and a plurality of second cutting paths are formed on the packaging module, and a plurality of packaging units are defined by the plurality of first cutting paths and the plurality of second cutting paths jointly within the packaging module; performing a pre-cutting process to cut form the top end of the packaging module along each of the first cutting paths down to the surface of the first adhesive layer, thereby to remove a part of the packaging module and to expose a part of the first adhesive layer; disposing at least a reinforcing structure on at least a part of the exposed surface of the first adhesive layer; electrically connecting a plurality of optoelectronic components to the corresponding packaging units; performing a singulation process to cut from a remaining portion of the top end of the packaging module along each of the second cutting paths down to the first adhesive layer; and removing the at least a reinforcing structure, the first adhesive layer and the carrier to form a plurality of the electronic packages, wherein each of the electronic packages comprises the packaging unit and the optoelectronic component.

[0008] In the aforementioned manufacturing method of an electronic package, the top surface of the packaging unit has a plurality of electrical connecting pads, the bottom surface of each of the optoelectronic components has a plurality of electrical connectors, and each of the electrical connectors is corresponding in position and electrically connecting to a corresponding one of the electrical connecting pads.

[0009] In the aforementioned manufacturing method of an electronic package, the optoelectronic component is subjected to a thermal compression process for the plurality of electrical connectors on the bottom surface of the optoelectronic component to penetrate a second adhesive layer and to be electrically connected to the plurality of electrical connecting pads.

[0010] In the aforementioned manufacturing method of an electronic package, the electrical connectors are solder balls.

[0011] In the aforementioned manufacturing method of an electronic package, the second adhesive layer is a Thermal Compression Non-Conductive Polymer (TCNCP) film.

[0012] In the aforementioned manufacturing method of an electronic package, the reinforcing structure comprises a Die Attached Film (DAF) disposed on the first adhesive layer and a reinforcement disposed on the DAF.

[0013] In the aforementioned manufacturing method of an electronic package, the reinforcement is a dummy die or a dummy silicon (dummy Si).

[0014] In the aforementioned manufacturing method of an electronic package, the reinforcing structure corresponds to and covers the exposed surface of the first adhesive layer after the pre-cutting process.

[0015] In the aforementioned manufacturing method of an electronic package, the reinforcing structure covers at least a part of the surface of the first adhesive layer exposed from the packaging module between any two adjacent ones of the second cutting paths after the pre-cutting process.

[0016] In the aforementioned manufacturing method of an electronic package, the optoelectronic component is a photonic IC (PIC).

[0017] In the aforementioned manufacturing method of an electronic package, the packaging module is a fan-out packaging module.

[0018] In the aforementioned manufacturing method of an electronic package, there are a plurality of semiconductor components in the packaging module, and each of the optoelectronic components is electrically connected to at least one of the semiconductor components respectively.
[0019] In the aforementioned manufacturing method of an electronic package, the semiconductor component is an electronic IC (EIC).
[0020] In the aforementioned manufacturing method of an electronic package, the carrier is a glass substrate.
[0021] By the implementation of the present disclosure, the electronic package can be manufactured by disposing a reinforcing structure on the exposed surface of the first adhesive layer after performing pre-cutting, even if the exposed surface of the first adhesive layer is imposed by the heat and compression in the subsequent processes, deformation such as bulge will not occur due to the reinforcing structure. Therefore, the structure of the packaging module, especially the packaging unit in the packaging module, will not be damaged. In the meantime, the reinforcing structure can also maintain the distances between and relative positions of the various parts of the packaging module after pre-cutting, so that even if the various parts of the packaging modules are subjected to heat and compression, displacement due to the melting or softening of the first adhesive layer underneath will not occur and the defects caused by position deviations in the subsequent processes, such as singulation, can be avoided. Therefore, the manufacturing yield of the electronic package can be improved effectively. In addition, the aforementioned manufacturing method does not require additional or new material or equipment, the existing materials and machines can be used to solve the technical problems, so there will be no substantial additional costs required.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
[0022] FIG. **1**A to FIG. **1**C are schematic views illustrating the manufacturing processes of an conventional semiconductor package.
[0023] FIG. **1**B-**2** is a schematic top view of FIG. **1**B-**1**.
[0024] FIG. **2**A-**1** to FIG. **2**E are schematic views of an embodiment for the manufacturing method of an electronic package of the present disclosure.
[0025] FIG. **2**A-**2**, FIG. **2**B-**2** and FIG. **2**C-**2** are schematic top views of FIG. **2**A-**1**, FIG. **2**B-**1** and FIG. **2**C-**1** respectively.
[0026] FIG. **2**C-**3** is a schematic top view of another embodiment of FIG. **2**C-**2**.
DETAILED DESCRIPTION
[0027] Implementations of the present disclosure are illustrated using the following embodiments. One of ordinary skill in the art can readily appreciate other advantages and technical effects of the present disclosure upon reading the content of this specification.
[0028] It should be noted that the structures, ratios, sizes, etc. shown in the drawings appended to this specification are to be construed in conjunction with the disclosure of this specification in order to facilitate understanding of those skilled in the art. They are not meant to limit the implementations of the present disclosure, and therefore have no substantial technical meaning. Any modifications of the structures, changes of the ratio relationships or adjustments of the sizes, are to be construed as falling within the range covered by the technical content disclosed herein to the extent of not causing changes in the technical effects created and the objectives achieved by the present disclosure. Meanwhile, terms such as “on,” “first,” “second,” “third,” “a,” and the like recited herein are for illustrative purposes, and are not meant to limit the scope in which the present disclosure can be implemented. Any variations or modifications to their relative relationships, without changes in the substantial technical content, should also to be regarded as within the scope

in which the present disclosure can be implemented.

[0029] FIG. **2**A-**1** to FIG. **2**E are schematic views of the manufacturing method of an electronic package **2** of the present disclosure, wherein FIG. **2**A-**2**, FIG. **2**B-**2** and FIG. **2**C-**2** are schematic partial top views of FIG. **2**A-**1**, FIG. **2**B-**1** and FIG. **2**C-**1** respectively.

[0030] As shown in FIG. **2**A-**1** and FIG. **2**A-**2**, a packaging module **20** is provided at first. The packaging module **20** is adhered to a surface of a carrier **30** by a first adhesive layer **40**. A plurality of first cutting paths **21***a* and a plurality of second cutting paths **21***b* are formed on the packaging module **20**. The first cutting paths **21***a* and second cutting paths **21***b* are formed along different directions respectively and intersecting to each other. In an embodiment, the plurality of first cutting paths **21***a* extend along the longitudinal direction, and the plurality of second cutting paths **21***b* extend along the transverse direction. The plurality of first cutting paths **21***a* and the plurality of second cutting paths **21***b* jointly define a plurality of packaging units **22** within the packaging module **20**. That is, each packaging unit **22** is surrounded and defined jointly by two adjacent ones of the first cutting paths **21***a* and two adjacent ones of the second cutting paths **21***b* in the packaging module **20**.

[0031] As shown in FIG. **2**B-**1** and FIG. **2**B-**2**, a pre-cutting process is performed. This process is performed to cut the packaging module **20** from the top end of the packaging module **20** along each of the first cutting paths **21***a* down to the surface of the first adhesive layer **40**. Thus, the part of the packaging module **20** originally located underneath each of the first cutting paths **21***a* is removed, and the part of the first adhesive layer **40** originally located beneath each of the first cutting paths **21***a* is exposed from the packaging module.

[0032] Then, as shown in FIG. **2**C-**1** and FIG. **2**C-**2**, a plurality of reinforcing structures **50** is disposing on at least a part of the exposed surface of the first adhesive layer **40** after performing the pre-cutting process. In other words, each of the reinforcing structures **50** is located between any two adjacent ones of the packaging units **22** and across over the first cutting path **21***a* separating the two adjacent ones of the packaging units **22**. Preferably, the two side ends of each of the reinforcing structures **50** abut the sides of the two adjacent ones of the packaging units **22**, respectively.

[0033] In FIG. **2**D, a plurality of the optoelectronic components **60** are adhered to the top surfaces of the packaging units **22** by a second adhesive layer **70**.

[0034] After each of the optoelectronic components **60** is adhered onto the corresponding one of the packaging units **22**, each of the optoelectronic components **60** is electrically connected to the corresponding one of the packaging units **22** by thermal compression or soldering.

[0035] Then, a singulation process is performed to cut the remaining portion of the packaging module **20** from its top end downward to the first adhesive layer **40** along the second cutting path **21***b* shown in FIG. **2**A-**2**. Accordingly, a plurality of the packaging units **22** are separated from each other. But at this moment, the packaging units **22** are still adhered to the first adhesive layer **40**.

[0036] Finally, the aforementioned reinforcing structures **50**, the first adhesive layer **40** and the carrier **30** are all removed to form a plurality of the electronic packages **2** as shown in FIG. **2**E. Each of the electronic packages **2** comprises a packaging unit **22** and an optoelectronic component **60**.

[0037] More specifically, the carrier **30** may be a substrate or a carrier board made of insulating material. For instance, a glass substrate may be served as the carrier **30**.

[0038] Referring to FIG. **2**A-**1**, the packaging module **20** in an embodiment may be a fan-out packaging module **20**, wherein a circuit structure **23** may be included therein. The circuit structure **23** comprises at least a circuit layer **231**, such as a fan-out redistribution layer (RDL). The packaging module **20** may also have a plurality of semiconductor components **221**. These semiconductor components **221** are configured in a way that each of the packaging units **22** in the packaging module **20** has at least one semiconductor component **221** therein. Each of the semiconductor components **221** may be an Electronic IC ("EIC" in short), or other semiconductor chips or electronic modules that meets the functional requirements when in design.

[0039] As shown in FIG. **2**C-**1**, the reinforcing structure **50** may include a die attached film (DAF) **51** disposed on the first adhesive layer **40** and a reinforcement element **52** disposed on the DAF **51**. Preferably, the reinforcement element **52** is a dummy die or dummy silicon (Si). Besides, the reinforcement element **52** may also be made of plastics, such as polycarbonate (PC), polyethylene (PE), polyoxymethylene (POM), polyetheretherketone (PEEK), other engineering plastics, or composite materials. As long as a material can meet the requirements for the manufacturing processes, it can be used to manufacture the reinforcement element **52**. Therefore, there is no additional limitation for implementation.

[0040] As shown in FIG. **2**C-**2**, in certain embodiments, the reinforcing structure **50** may be designed to correspondingly cover the exposed surface of the first adhesive layer **40** after performing the pre-cutting process. That is, each of the reinforcing structures **50** is designed and made in a strip shape, and is disposed on the first adhesive layer **40** along each of the original first cutting paths **21***a*, so as to correspondingly cover all of the exposed surfaces of the first adhesive layer **40**. The advantage for this design is that it can provide desired structural reinforcement effect. On the other hand, this design will also require the subsequent singulation process to cut the reinforcing structure **50** lying across each of the second cutting paths **21***b* at the same time.

[0041] In another embodiment shown in FIG. **2**C-**3**, the reinforcing structure **50** is merely designed to cover a part of the exposed surface of the first adhesive layer **40** after performing the pre-cutting process. In other words, the part of the exposed surface of the first adhesive layer **40** underneath the first cutting paths **21***a* and located between any two adjacent ones of the second cutting paths **21***b*. This arrangement requires the reinforcing structure **50** to be disposed on the part of the first adhesive layer **40** that is easily to bulge and/or the part of packaging module **20** that is easily to dislocate at the time the first adhesive layer **40** is subjected to subsequent thermal compression process. As a result, this design can not only provide sufficient structural reinforcement effect, but also can save the energy consumption and decrease the depletion of cutting equipment, as well as save the process time.

[0042] In addition, a plurality of electrical connecting pads **24** are disposed on the top surface of each of the packaging units **22** in the packaging module **20** for the subsequent electrical connection with the optoelectronic component **60**. As for the second adhesive layer **70** formed on the packaging module **20**, it can be a thermal compression non-conductive polymer (“TCNCP” in short) film or other adhesive material that can meet the requirements of the manufacturing processes.

[0043] Each of the optoelectronic components **60** is electrically connected to at least one of the semiconductor component **221** disposed in the corresponding packaging module **20**. Therefore, there are a plurality of electrical connectors **61** provided on the bottom surface of each of the optoelectronic components **60**. The electrical connectors **61** are solder balls, for instance, or other types of conductive bumps. Each of the electrical connectors **61** is corresponding in position to an electrical connecting pad **24** on the packaging module **20**, and is electrically connected to a corresponding one of the electrical connecting pads **24** after the second adhesive layer **70** is melted or softened by thermal compression to allow the optoelectronic components **60** to embed in the second adhesive layer **70**. Thereby, each of the optoelectronic components **60** is electrically connected with a corresponding one of the packaging units **22**.

[0044] In an embodiment, the optoelectronic component **60** is a photonic IC (“PIC” in short). But the optoelectronic component **60** may be other types of optoelectronic component that can meet the functional requirements.

[0045] In summary, in the manufacturing method of an electronic package of the present disclosure, the reinforcement structure is disposed on the exposed surface of the first adhesive layer after the pre-cutting process is performed, so even if the first adhesive layer is imposed by the thermal compression in the subsequent processes, deformation such as bulge will not occur on the surface of the first adhesive layer due to the reinforcing structure. Therefore, the structure of the packaging module will not be damaged, especially the packaging units. In the meantime, the reinforcing structure can also maintain the distances and relative positions between the packaging units of the packaging module after the pre-cutting process, such that, even if the packaging module is subjected to thermal compression, displacement due to the melting or softening of the first adhesive layer underneath will not occur and thus can be kept on the desired positions to avoid the defects caused by position deviations in subsequent processes, such as singulation. Therefore, the manufacturing yield of the electronic package can be improved effectively. In addition, the aforementioned manufacturing method does not require additional or new material or equipment, existing materials and equipment can be well used to solve the technical problems, and there will be no substantial additional costs produced.

[0046] The above embodiments are set forth to illustrate the principles of the present disclosure, and should not be interpreted as to limit the present disclosure. The above embodiments can be modified by one of ordinary skill in the art without departing from the scope of the present disclosure as defined in the appended claims. Therefore, the scope of protection of the right of the present disclosure should be listed as the following appended claims.

Claims

1. A manufacturing method of an electronic package, comprising: adhering a packaging module onto a surface of a carrier through a first adhesive layer, wherein a plurality of first cutting paths and a plurality of second cutting paths are formed on the packaging module, and a plurality of packaging units are defined by the plurality of first cutting paths and the plurality of second cutting paths jointly within the packaging module; performing a pre-cutting process to cut from a top end of the packaging module along each of the first cutting paths down to a surface of the first adhesive layer, to thereby remove a part of the packaging module and to expose a part of the first adhesive layer; disposing at least a reinforcing structure on at least a part of the exposed surface of the first adhesive layer; electrically connecting a plurality of optoelectronic components to the corresponding packaging units; performing a singulation process to cut from a remaining portion of the top end of the packaging module along each of the second cutting paths down to the first adhesive layer; and removing the at least a reinforcing structure, the first adhesive layer and the carrier to form a plurality of electronic packages, wherein each of the electronic packages comprises the packaging unit and the optoelectronic component.

2. The manufacturing method of claim 1, wherein a top surface of each of the packaging units has a plurality of electrical connecting pads, a bottom surface of each of the optoelectronic components has a plurality of electrical connectors, and each of the electrical connectors is corresponding in position and electrically connected to a corresponding one of the electrical connecting pads.

3. The manufacturing method of claim 2, wherein the optoelectronic component is subjected to a thermal compression process for the plurality of electrical connectors on the bottom surface of the optoelectronic component to penetrate a second adhesive layer, so as for the electrical connectors to be electrically connected to the plurality of electrical connecting pads.

4. The manufacturing method of claim 3, wherein the second adhesive layer is a Thermal Compression Non-Conductive Polymer (TCNCP) film.

5. The manufacturing method of claim 1, wherein the reinforcing structure comprises a Die Attached Film (DAF) disposed on the first adhesive layer and a reinforcement element disposed on the DAF.

6. The manufacturing method of claim 5, wherein the reinforcement element is a dummy die or a dummy silicon (dummy Si).

7. The manufacturing method of claim 1, wherein the reinforcing structure corresponds in position to and covers the exposed surface of the first adhesive layer after the pre-cutting process.

8. The manufacturing method of claim 1, wherein the reinforcing structure covers at least a part of the surface of the first adhesive layer exposed from the packaging module between any two adjacent ones of the second cutting paths after the pre-cutting process.
9. The manufacturing method of claim 1, wherein the optoelectronic component is a photonic IC (PIC).
10. The manufacturing method of claim 1, wherein the packaging module is a fan-out packaging module.
11. The manufacturing method of claim 1, wherein there are a plurality of semiconductor components in the packaging module, and each of the optoelectronic components is electrically connected to at least one of the semiconductor components respectively.
12. The manufacturing method of claim 11, wherein the semiconductor component is an electronic IC (EIC).
13. The manufacturing method of claim 1, wherein the carrier is a glass substrate.